

Statistics

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1 Probability

1. [Introduction, Definition of Probability](#)
2. [Sampling, Combinatorics](#)
3. [Uniform Probability, Basic Properties of Probability](#)
4. [Conditional Probability](#)
5. [Bayes' Rule, Independence](#)
6. [Random Variables](#)
7. [Probability Distributions](#)
8. [Probability Distributions](#)
9. [Independent Trials and Sampling, Binomial, Geometric, and Poisson Distributions](#)
10. [Binomial, Geometric, and Poisson Distributions](#)
11. [Expected Value](#)
12. [Review, Evening Midterm](#)
13. [Variance, Normal \(Gaussian\) Distribution](#)
14. [Normal \(Gaussian\) Distribution, Normal Approximation](#)
15. [Normal Approximation](#)
16. [Confidence Intervals](#)
17. [Poisson Approximation, Exponential Distribution](#)
18. [Poisson Process](#)
19. [Moment Generating Function](#)
20. [Moment Generating Function](#)
21. [Functions of Random Variables, Joint Distributions](#)
22. [Joint Distributions, Independence of Random Variables](#)
23. [Review, Evening Midterm](#)
24. [Joint Distributions, Independence of Random Variables](#)

- 25. [Expectations of sums](#)
- 26. [Covariance, correlation, and variance of sums](#)
- 27. [Law of Large Numbers, Central Limit Theorem](#)
- 28. [Review](#)
- 29. [Review](#)

2 Survival Analysis

1. Introduction
2. Maximum Likelihood example
3. More intuition on Maximum Likelihood
4. Kaplan Meier & Nelson Aalen
5. Mean vs. Median vs. Restricted Mean
6. Cox Proportional Hazards
7. Cox PH: Partial Likelihood
8. Cox PH: Breslow Estimator
9. Accelerated Failure Time
10. AFT vs. Cox PH
11. Cox Extensions
 - (a) Stratified Cox
 - (b) Extended Cox
 - (c) Clustered Cox

3 Markov Processes

1. [Stochastic Processes, Conditional Probability, and Introduction to Markov Chains](#)
2. [Markov Chains: Conditional Probability, Transition Matrices, and Chapman-Kolmogorov Equations](#)
3. [Markov Chain Simulation, Chapman-Kolmogorov Equations, and Absorbing States](#)
4. [Markov Chains: Limiting Distributions, Communication Classes, Recurrence, and Hitting Times](#)
5. [Markov Chains: Geometric Distribution, Recurrence, Transience, and Random Walk Analysis](#)
6. [Markov Chains: Random Walks, Recurrence, Hitting Times, and Periodicity](#)
7. [Markov Chains: Aperiodicity, Limiting Distributions, Stationarity, and Eigenvectors](#)
8. [Markov Chains: Wald's Identity, Recurrence, Hitting Probabilities, and Expected Visits](#)
9. [First-Step Analysis: Hitting Times, Return Times, Visits, and Pattern Waiting](#)
10. [Continuous-Time Processes: Exponential Distribution, Memorylessness, Simulation, and Gamma Distribution](#)
11. [text](#)
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4 Time Series Analysis

1. [Noise Processes](#)
2. [Autocovariance](#)
3. [Estimation and Linear Models](#)
4. [Smoothing Methods](#)
5. [AR and MA Theory](#)
6. [ARMA and ARIMA](#)
7. [Autocorrelation and Stationarity](#)
8. [Stationarity and ACF in R](#)
9. [ACF and PACF, part 1](#)
10. [ACF and PACF, Part 2](#)
11. [Estimation for AR\(p\)](#)
12. [Forecasting for AR\(p\)](#)
13. [Durbin-Levinson and Innovations Algorithms](#)
14. [Estimation for ARMA](#)
15. [Forecasting for ARMA](#)
16. [Seasonal ARIMA Models](#)
17. [Seasonal ARIMA in R Studio](#)
18. [Periodic Processes and the DFT](#)
19. [Spectral Density and Distribution](#)
20. [Estimating the Spectral Density](#)
21. [Estimating the Spectral Density II](#)
22. [Estimating the Spectral Density III](#)
23. [Exponential Smoothing](#)
24. [The GARCH Process](#)
25. [TSA Review Lecture](#)

5 Monte Carlo Methods

6 Mathematical Statistics

1. Course introduction and fundamentals of random variables and probability distributions
2. Expected values, functions of random variables, and joint distributions
3. Dependence, indicator techniques, random variable transformations, and Gamma distribution
4. LOTUS, Gamma and Beta distributions, transformations, and distribution relationships
5. Non-invertible transformations, order statistics, moments, and moment generating functions
6. MGFs, moments, IID sums, distribution identification, and estimation basics
7. Expectation, variance, covariance, estimators, unbiasedness, and properties of sample means
8. Estimator comparison, MSE, bias-variance tradeoff, and exponential parameter estimation
9. MSE, asymptotic unbiasedness, convergence, inequalities, and weak law of large numbers
10. Consistency, useful convergence theorems, examples, and probability convergence properties
11. (a) Continuous mapping, joint convergence, sample variance, and convergence in distribution
(b) Convergence in probability implies convergence in distribution, but not conversely.
12. Convergence concepts, continuous mapping, Slutsky's theorem, and MGF methods
13. Asymptotic normality, Central Limit Theorem, normal approximations, chi-squared distribution, applications
14. Delta method, asymptotic normality, transformations, and sample variance distributions
15. Sample variance distributions, chi-squared, t-distribution, and confidence interval basics
16. Degrees of freedom, confidence intervals, normal and large-sample inference
17. Confidence intervals for exponential rate, variance, and mean differences
18. Two-sample confidence intervals, variance assumptions, hypothesis testing, and error types
19. Hypothesis testing errors, significance level, test size, power, and examples
20. Composite hypothesis testing, Type I/II errors, power functions, and test comparison
21. Hypothesis testing, two-tailed tests, variance parameters, and method-of-moments estimation
22. Maximum likelihood estimation, formal definitions, and examples across common distributions

23. MLE examples, invariance property, Pareto estimation, and Cramér-Rao lower bound
24. Cramér-Rao bound, Fisher information, MLE properties, efficiency, and Jensen's inequality
25. Jensen's inequality, MLE properties, consistency, asymptotic unbiasedness, and examples
26. Efficiency, asymptotic efficiency, asymptotic normality, and MLE examples
27. MLE inference, CRLB, hypothesis testing, UMVUE, and sufficient statistics
28. Sufficient statistics, factorization criterion, conditional moments, and Rao-Blackwell theorem
29. Rao-Blackwell theorem, variance calculations, complete statistics, and uniqueness
30. UMVUEs, Lehmann-Scheffé theorem, exponential families, and distribution examples
31. UMVUEs, Rao-Blackwell, minimal sufficiency, Bahadur's theorem, likelihood ratio criterion
32. Completeness, minimal sufficiency, ancillary statistics, location-scale invariance, and independence
33. Basu's Theorem and an Introduction to Generalized Hypothesis Testing
34. Best and Uniformly Most Powerful Hypothesis Tests
35. Uniformly Most Powerful Hypothesis Tests
36. Generalized Likelihood Ratio Tests
37. Generalized Likelihood Ratio Tests (asymptotics!)

7 Multivariate and Regression Analysis

8 Graphical Models and Causal Inference

1. [Overview](#)
2. [Undirected graphical models - definitions](#)
3. [Conditional independence axioms](#)
4. [Equivalence of definitions, Gaussian distributions](#)
5. [Properties of Gaussians, Maximum likelihood estimation](#)
6. [Maximum Likelihood Estimation for Gaussian graphical models](#)
7. [MLE for Gaussian graphical models; Chordal graphs](#)
8. [Learning the graph for Gaussian graphical models, Discrete graphical models](#)
9. [Maximum Likelihood Estimation for Log-linear models](#)
10. [Learning Undirected Graphs](#)
11. [Learning Undirected Graphs; Introduction to causality](#)
12. [Directed graphical models](#)
13. [Markov equivalence. Directed Gaussian graphical models](#)
14. [Directed Gaussian graphical models](#)
15. [Learning directed acyclic graphs from observations](#)
16. [Structural equation models and interventions](#)
17. [Identification of causal effects from observations](#)
18. [Counterfactuals, Instrumental variables, and Linear Causal Models](#)

9 High-Dimensional Probability and Applications in Data Science

1. Curse of Dimensionality and Monte Carlo Methods in High Dimensions
2. Convexity and Carathéodory Theorems via Maurey's Empirical Probabilistic Method
3. Approximate Carathéodory: Portfolios, Factor Analysis, Covering Numbers, and Polytopes
4. High-Dimensional Geometry: Polytope Volumes, Hyperbolic Intuition, and Gaussian Concentration
5. Concentration of Independent Sums: Markov, Chebyshev, Berry-Esseen, and Hoeffding
6. Hoeffding's Inequality and Robust Mean Estimation via Median-of-Means
7. Concentration Inequalities: Hoeffding Variants, Poisson Limits, and Chernoff Bounds
8. Random Graphs: Deviations, Erdős-Rényi Models, Phase Transitions, and Regularity
9. Sparse Random Graph Irregularity, Probabilistic Guarantees, and Geometric Discrepancy
10. Discrepancy Theory: Proof and Key Ideas Behind the Discrepancy Theorem
11. Spaces of Random Variables: Norms, L_p Spaces, and Fundamental Inequalities
12. Orlicz Spaces and Fundamental Properties of Subgaussian Random Variables
13. Subgaussian Distributions: Equivalent Properties and the Subgaussian Norm
14. Subgaussian and Subexponential Concentration: Hoeffding and Bernstein Inequalities
15. Thin Shell Phenomenon and Dimension Reduction via Johnson-Lindenstrauss Lemma
16. Combinatorial Optimization: Ising Models, Clustering, Spectral and Semidefinite Relaxations
17. Gram Matrices, Semidefinite Relaxation, and the Goemans-Williamson Max-Cut Algorithm
18. Grothendieck's Inequality, Tensor Calculus, and Krivine's Bound
19. Binary Classification, Support Vector Machines, Kernel Trick, and RBF Kernels
20. SVMs, Kernel Methods, Neural Networks, and Neural Tangent Kernels
21. Linear Algebra Review: Spectral Decomposition, Singular Values, and Frobenius Norm
22. High-Dimensional Random Vectors, Covariance, Operator Norms, and Principal Component Analysis
23. Covariance Estimation via Sample Matrices, Stochastic Processes, Brownian Motion, and Epsilon-Nets

24. Covariance Estimation, Spectral Perturbation, and PCA via Weyl and Davis-Kahan
25. Random Matrices: Wigner's Semicircle Law via the Stieltjes Transform
26. Marchenko-Pastur Law
27. Random Matrix Theory: Spectral Laws, Universality, Spikes, and Eigenvalue Statistics
28. Functional Calculus, Loewner Order, and Matrix Monotone Functions
29. Lieb's Trace Inequality and Matrix Hoeffding Concentration Bounds
30. Matrix Bernstein Inequality and Community Detection in Stochastic Block Models
31. Spectral Community Recovery in Networks: Algorithms and Stochastic Block Model Guarantees
32. Community Detection, Stochastic Block Models, and Modern Dimensionality Reduction Visualization
33. Low-Dimensional Learning: Effective Rank, Dimension, and Covariance Estimation
34. Foundations of Supervised Learning: Risk, Overfitting, ERM, and Generalization
35. Generalization Bounds and VC Dimensions of Classical Classifier Families
36. VC Theory: Pajor's Lemma and the Sauer-Shelah Lemma
37. Sauer-Shelah Lemma, Hyperplane Arrangements, Empirical Processes, and Uniform Laws
38. Uniform Laws via Symmetrization, Rademacher Complexity, and VC Generalization Bounds
39. Gaussian and Subgaussian Processes: Dudley's Integral Inequality and Bounds
40. Dudley's Inequality, Uniform Laws, and Lipschitz Regression Applications
41. Isoperimetric Inequalities and Gaussian Concentration for Lipschitz Functions

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